University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2021, Part 2

Answer four problems. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Prove the Chinese remainder theorem: Let R be a commutative ring with 1 and let I, J be ideals in R such that $I + J = R$. Then $IJ = I \cap J$ and there is a ring isomorphism $R/IJ \cong (R/I) \times (R/J)$.

2. Prove that the ring $\mathbb{Z}[\sqrt{-2}] = \{a + b\sqrt{-2} : a, b \in \mathbb{Z}\}$ is a Euclidean domain with respect to the field norm

$$N : \mathbb{Z}[\sqrt{-2}] \to \mathbb{Z}_{\geq 0}, \qquad N(a + b\sqrt{-2}) = a^2 + 2b^2.$$

3. Prove that each of the following is irreducible in the indicated ring.

(a) $2X^3 + 2X^2 + 3X + 1 \in \mathbb{Q}[X]$

(b) $Y^3 + X^2 + XY - Y \in \mathbb{Q}[X, Y]$

(c) $X^4 - X^3 + 4X^2 - 15 \in \mathbb{Z}[X]$

4. Let F be a field and let V be a vector space over F. Let $S \subset V$ be a set which spans V. Use Zorn's lemma to prove that V has a basis which is contained in S.

5. Determine the number of similarity classes of matrices $A \in \mathrm{M}_5(\mathbb{C})$ satisfying $A^3 = 0$. Give a representative for each similarity class in Jordan canonical form.

6. This problem has two parts.

(a) Let L/K be a field extension of degree 5 and let $f(X) \in K[X]$ have degree 4. Prove that if $\alpha \in L$ is a root of $f(X)$, then $\alpha \in K$.

(b) Give an example of a field extension L/K of degree 4, a polynomial $f(X) \in K[X]$ of degree 5, and a root α of $f(X)$ such that $\alpha \in L$ but $\alpha \notin K$.